

MAURÍCIO H. VANCINE

Ecologist, Data Scientist, and PhD in Ecology

State University of Campinas (Unicamp), Campinas, SP, Brazil

Institute of Biology, Department of Animal Biology, Rua Monteiro Lobato, 255, 13083862

CAREER SUMMARY

Environmental scientist and ecologist specializing in biodiversity conservation, spatial ecology, and species distribution modeling. My research integrates ecological theory, geospatial analysis, and machine learning to investigate how landscape structure and climate change affect biodiversity patterns and ecosystem resilience.

I develop computational tools for ecological analysis and actively contribute to teaching and training in the R programming language.

Currently, I am a Postdoctoral researcher at [LDDiv](#) at [Unicamp](#) where I integrate landscape metrics and species distribution modeling (SDM) to assess the impacts of edge effects, fires, and climate change on the distribution and interactions of animals and plants across multiple spatial scales in the Amazon.

ACADEMIC EXPERTISE

Spatial Ecology, Species Distribution Modeling, Landscape Ecology, Geospatial Analysis (R, GIS), Machine Learning for Ecology, Ecological Data Analysis (R).

RESEARCH INTERESTS

Habitat loss and fragmentation, landscape metrics, species distribution modeling (SDM), community ecology, amphibian ecology, conservation biology, teaching the R programming language and statistics.

EDUCATION

São Paulo State University (UNESP)

Mar 2011 - Mar 2015

B.S. in Ecology

GPA: 3.43

Thesis: Effect of fragmentation on the persistence of anuran amphibians (Amphibia: Anura) within the Atlantic Forest

Library: [link](#)

Advisor: Prof. Dr. Milton Cezar Ribeiro

São Paulo State University (UNESP)

Mar 2016 - Jul 2018

M.S., Graduate Program in Biological Science (Zoology)

GPA: 4.00

Thesis: Diversity, distribution and effect of climate change on Atlantic Forest amphibian communities

Library: [link](#)

Advisor: Prof. Dr. Célio Fernando Baptista Haddad

São Paulo State University (UNESP)

Mar 2020 - Jul 2024

Ph.D., Graduate Program in Ecology, Evolution and Biodiversity

GPA: 4.00

Dissertation: Landscape structure as a predictor of taxonomic and functional diversity of amphibians in the Atlantic Forest

Library: [link](#)

Advisor: Prof. Dr. Milton Cezar Ribeiro

COMPLEMENTARY EDUCATION

- **Ecological Data Analysis with R (40 h)** 2011
São Paulo State University (UNESP), Rio Claro, SP, Brazil
- **Biology and conservation of amphibians and reptiles (44 h)** 2015
Instituto Boitatá, IBEC, Alfenas, MG, Brazil, [link](#)
- **V Southern-Summer School on Mathematical Biology (53 h)** 2016
São Paulo State University (UNESP), São Paulo, SP, Brazil, [link](#)
- **Geoprocessing with GRASS GIS (24 h)** 2016
São Paulo State University (UNESP), Rio Claro, SP, Brazil
- **Introduction to Hierarchical Modeling (45 h)** 2019
Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, RS, Brazil, [link](#)
- **School on Community Ecology: from patterns to principles (60 h)** 2020
São Paulo State University (UNESP), São Paulo, SP, Brazil, [link](#)
- **Hierarchical Modelling of Species Communities with the R-package Hmsc (25 h)** 2020
University of Helsinki (Online), Helsinki, Finland, [link](#)
- **Joint Species Distribution Modelling with HMSC (45 h)** 2022
University of Jyväskylä (Online), Jyväskylä, Finland, [link](#)
- **Data Science Journey (330 h)** 2022
Ômega Data Science (Online), Curitiba, PR, Brazil, [link](#)
- **English writing for researchers (30 h)** 2023
Una Assessoria Linguística (Online), Uberlândia, Minas Gerais, Brazil, [link](#)
- **Geospatial Analysis with Python (50 h)** 2026
AmbGEO (Online), Florianópolis, SC, Brazil, [link](#)
- **Google Earth Engine (GEE) (50 h)** 2026
AmbGEO (Online), Florianópolis, SC, Brazil, [link](#)

PROFESSIONAL EXPERIENCE

- **Intern (300 h)** 2014
Role: Compulsory curricular internship focused on Species Distribution Modeling (SDM)
Institution: Luiz de Queiroz College of Agriculture, University of São Paulo (USP), Ecology, Management and Conservation of Wild Fauna Lab (LEMaC)
Supervisor: Prof. Dr. Katia Maria Paschoaletto Micchi de Barros Ferraz
- **Research assistant (2000 h)** 2015-2016
Role: Statistical and spatial analyses of mammal and ants biodiversity in the Atlantic Forest biome
Institution: São Paulo State University (UNESP), Spatial Ecology and Conservation Lab (LEEC)
Supervisor: Prof. Dr. Milton Cezar Ribeiro
- **Postdoctoral researcher** 2024-interrupted
Role: Impacts of edge effects, fire, and climate change on vegetation composition and functional diversity across spatial scales in the Amazon rainforest
Institution: State University of Campinas (Unicamp), LDDiv
Supervisor: Prof. Dr. Mathias Mistretta Pires

- **Postdoctoral researcher fellowship** 2026-present
Role: Effects of habitat degradation on inferring the local dominance of tree species in the Amazon rainforest
Institution: Yale School of the Environment, Brando's Lab
Supervisor: Prof. Dr. Paulo Brando

RESEARCH EXPERIENCE

- **Undergraduate research** 2013-2015
Title: Effect of fragmentation on the persistence of anuran amphibians (Amphibia: Anura) within the Atlantic Forest
Aim: To evaluate the relative contribution of landscape indices (percentage of forest cover, connectivity, relief and urban proximity) to the persistence of species using species distribution modeling.
- **Master's research** 2016-2018
Title: Effect of Landscape Modifications and Climate Changes on the Persistence of Amphibians in the Atlantic Forest
Aim: To present an assessment of the surveys of the amphibian communities for the Atlantic Forest; to investigate how habitat loss and fragmentation affect the persistence of amphibians; and to investigate how climate change affect the future persistence of the genus *Brachycephalus*.
- **Doctoral research** 2020-2024
Title: Landscape structure as a predictor of taxonomic and functional diversity of amphibians in the Atlantic Forest
Aim: To analyze the structure of landscapes throughout the Atlantic Forest; to analyze how landscape structure affects the taxonomic and functional diversity of amphibian communities; and to analyze this same question at the species-specific level, in addition to analyzing co-occurrences using JSDMs.
- **Postdoctoral research** 2024-interrupted
Title: Impacts of edge effects, fire, and climate change on vegetation composition and functional diversity across spatial scales in the Amazon rainforest
Aim: To understand how edge effects, fires, and climate change acting at local scales can influence the distribution and composition patterns of vegetation at different spatial scales in the Amazon biome, also affecting the potential distribution of vertebrate fauna. In addition, we will investigate how plant functional attributes respond to these factors, with implications for the functional diversity of plant and animal communities.
- **Postdoctoral research** 2026-present
Title: Effects of habitat degradation on inferring the local dominance of tree species in the Amazon rainforest
Aim: To predict how tree species respond to local disturbances-including edge effects, fires, droughts, and windthrow events-by incorporating habitat degradation into models of their regional potential distribution across the Amazon.

FUNDING

- **Undergraduate Research Scholarship**, São Paulo Research Foundation (FAPESP) 2013-2015
Scholarship: [2013/02883-7](#)
Value: BRL 10,539.87

- **Master's Scholarship**, São Paulo Research Foundation (FAPESP) 2017-2018
Scholarship: [2017/09676-8](#)
Value: BRL 16,248.54
- **Doctoral Scholarship**, São Paulo Research Foundation (FAPESP) 2022-2024
Scholarship: [2022/01899-6](#)
Value: BRL 91,002.24
- **Postdoctoral Scholarship**, São Paulo Research Foundation (FAPESP) and U.S. National Science Foundation (NSF) 2024-interrupted
Scholarship: [2024/19865-6](#)
Value: BRL 316,800.00
- **Postdoctoral researcher fellowship**, São Paulo Research Foundation (FAPESP) and U.S. National Science Foundation (NSF) 2026-present
Fellowship: [2025/15581-6](#)
Value: BRL 55,070.00 (FAPESP/NSF), USD 32,604.00 (FAPESP/NSF), and USD 1,602.00 (Yale)

AWARDS AND HONORS

- High Academic Performance Award, São Paulo State University (UNESP) 2015
- The Harry R. Painton Award, American Ornithological Society 2023

PUBLICATIONS

Peer-reviewed (most relevant)

1. Vancine MH, Muylaert RL, Niebuhr BB, Oshima JEF, Tonetti V, Bernardo R, De Angelo C, Rosa MR, Grohmann CH, Ribeiro MC. 2024. **The Atlantic Forest of South America: spatiotemporal dynamics of remaining vegetation and implications for conservation.** *Biological Conservation* 291:110499. [10.1016/j.biocon.2024.110499](#)
2. Gonçalves-Souza T, Chase J, Haddad N, Vancine MH, Melo FPL, Aizen M, Bernard E, Chiarello GA, Didham R, Faria D, Gibb H, Lima M, Magnago L, Mariano Neto E, Nogueira A, Nemésio A, Passamani M, Pinho BX, Rocha-Santos L, Rodrigues R, Safar N, Santos B, Soto-Werschitz A, Tabarelli M, Uehara-Prado M, Vasconcelos H, Vieira S, Sanders N. 2025. **Species turnover does not rescue biodiversity in fragmented landscapes.** *Nature* 640:702-706. [10.1038/s41586-025-08688-7](#)
3. Beca G, Vancine MH, Carvalho CS, Pedrosa F, Alves RSC, Buscariol D, Peres CA, Ribeiro MC, Galetti M. 2017. **High mammal species turnover in forest patches immersed in biofuel plantations.** *Biological Conservation* 210:352-359. [10.1016/j.biocon.2017.02.033](#)

Peer-reviewed (complete list - May 27, 2026)

1. Beca G, Vancine MH, Carvalho CS, Pedrosa F, Alves RSC, Buscariol D, Peres CA, Ribeiro MC, Galetti M. 2017. **High mammal species turnover in forest patches immersed in biofuel plantations.** *Biological Conservation* 210:352-359. [10.1016/j.biocon.2017.02.033](#)
2. de Castro Pena JC, Goulart F, Fernandes GW, Hoffmann D, Leite FS, dos Santos NB, Soares-Filho B, Sobral-Souza T, Vancine MH, Rodrigues M. 2017. **Impacts of mining activities on the potential geographic distribution of eastern Brazil mountaintop endemic species.** *Perspectives in Ecology and Conservation* 15(3):172-178. [10.1016/j.pecon.2017.07.005](#)

3. Regolin AL, Cherem JJ, Graipel ME, Bogoni JA, Ribeiro JW, **Vancine MH**, Castilho PVD. 2017. **Forest cover influences occurrence of mammalian carnivores within Brazilian Atlantic Forest.** *Journal of Mammalogy* 98(6):1721-1731. [10.1093/jmammal/gyx103](https://doi.org/10.1093/jmammal/gyx103)
4. Sobral-Souza T, **Vancine MH**, Ribeiro MC, Lima-Ribeiro MS. 2018. **Efficiency of protected areas in Amazon and Atlantic Forest conservation: A spatio-temporal view.** *Acta Oecologica* 87:1-7. [10.1016/j.actao.2018.01.001](https://doi.org/10.1016/j.actao.2018.01.001)
5. Muylaert RL, **Vancine MH**, Bernardo R, Oshima JEF, Sobral-Souza T, Tonetti VR, Ribeiro MC. 2018. **Uma nota sobre os limites territoriais da Mata Atlântica.** *Oecologia Australis* 22(3):302-311. [10.4257/oeco.2018.2203.09](https://doi.org/10.4257/oeco.2018.2203.09)
6. **Vancine MH**, Duarte KS, Souza YS, Giovanelli JGR, Sobrinho PMM, López A, Bovo RP, Maffei F, Lion MB, Ribeiro-Júnior JW, Brassaloti R, Ortiz C, Sawakuchi HO, Forti LR, Cacciali P, Bertoluci J, Haddad CFB, Ribeiro MC. 2018. **ATLANTIC AMPHIBIANS: a data set of amphibian communities from the Atlantic Forests of South America.** *Ecology [Data Paper]* 99(7):1692-1692. [10.1002/ecy.2392](https://doi.org/10.1002/ecy.2392)
7. Ferro e Silva AM, Sobral-Souza T, **Vancine MH**, Muylaert RL, Abreu AP, Pelloso SM, Carvalho MDB, Andrade L, Ribeiro MC, Toledo MJ. 2018. **Spatial prediction of risk areas for vector transmission of *Trypanosoma cruzi* in the State of Paraná, southern Brazil.** *PLoS Neglected Tropical Diseases* 12(10):e0006907. [10.1371/journal.pntd.0006907](https://doi.org/10.1371/journal.pntd.0006907)
8. Bertassoni A, Costa RT, Gouvea JA, Bianchi RDC, Ribeiro JW, **Vancine MH**, Ribeiro MC. 2019. **Land-use changes and the expansion of biofuel crops threaten the giant anteater in south-eastern Brazil.** *Journal of Mammalogy* 100(2):435-444. [10.1093/jmammal/gyz042](https://doi.org/10.1093/jmammal/gyz042)
9. Moraes AM, **Vancine MH**, Moraes AM, Oliveira Cordeiro CL, Pinto MP, Lima AA, Sobral-Souza T. 2019. **Predicting the potential hybridization zones between native and invasive marmosets within Neotropical biodiversity hotspots.** *Global Ecology and Conservation* 20:e00706.06. [10.1016/j.gecco.2019.e00706](https://doi.org/10.1016/j.gecco.2019.e00706)
10. Santos JP, SobralSouza T, Brown Jr KS, **Vancine MH**, Ribeiro MC, Freitas AV. 2020. **Effects of landscape modification on species richness patterns of fruitfeeding butterflies in Brazilian Atlantic Forest.** *Diversity and Distributions* 26(2):196-208. [10.1111/ddi.13007](https://doi.org/10.1111/ddi.13007)
11. Marjakangas E, Abrego N, Grøtan V, Lima RAF, Bello C, Bovendorp RS, Culot L, Hasui E, Muylaert RL, Lima F, Niebuhr B, Oliveira AA, Pereira L, Prado I, Stevens RD, **Vancine MH**, Ribeiro MC, Galetti M, Ovaskainen O. 2020. **Fragmented tropical forests lose mutualistic plant-animal interactions.** *Diversity and Distributions* 26(2):154-168. [10.1111/ddi.13010](https://doi.org/10.1111/ddi.13010)
12. Bello C, Cintra ALP, Barreto E, **Vancine MH**, Sobral-Souza T, Graham CH, Galetti M. 2021. **Environmental niche and functional role similarity between invasive and native palms in the Atlantic Forest.** *Biological Invasions* 23(3):741-754. [10.1007/s10530-020-02400-8](https://doi.org/10.1007/s10530-020-02400-8)
13. Gusmão RA, Hernandez FA, **Vancine MH**, Naka LN, Doña J, GonçalvesSouza T. 2021. **Host diversity outperforms climate as a global driver of symbiont diversity in the birdfeather mite system.** *Diversity and Distributions* 27(3):416-426. [10.1111/ddi.13201](https://doi.org/10.1111/ddi.13201)
14. Da Silveira NS, **Vancine MH**, Jahn AE, Pizo MA, Sobral-Souza T. 2021. **Future climate change will impact the size and location of breeding and wintering areas of migratory thrushes in South America.** *The Condor* 123(2):duab006. [10.1093/ornithapp/duab006](https://doi.org/10.1093/ornithapp/duab006)
15. Jover A, Cabrera A, Ramos A, **Vancine MH**, Suárez AM, Machell J, Pérez-Lloréns JL. 2021. **Distribution of macroalgae epiphytes and host species from the Cuban marine shelf inferred**

- from ecological modelling. *Aquatic Botany* 172:103395. [10.1016/j.aquabot.2021.103395](https://doi.org/10.1016/j.aquabot.2021.103395)
16. Bercê W, Bello C, Mendes CP, Vancine MH, Galetti M, Ballari SA. 2021. **Invasive wild boars distribution overlap with threatened native ungulate in Patagonia.** *Journal of Mammalogy* 102(5):1298-1308. [10.1093/jmammal/gyab099](https://doi.org/10.1093/jmammal/gyab099)
 17. Oshima JEF, Jorge MLS, Sobral-Souza T, Börger L, Keuroghlian A, Peres CA, Vancine MH, Col-
leni B, Ribeiro MC. 2021. **Setting priority conservation management regions to reverse rapid
range decline of a key neotropical forest ungulate.** *Global Ecology and Conservation* 31:e01796.
[10.1016/j.gecco.2021.e01796](https://doi.org/10.1016/j.gecco.2021.e01796)
 18. Monteiro ECS, Pizo MA, Vancine MH, Ribeiro MC. 2022. **Forest cover and connectivity have
pervasive effects on the maintenance of evolutionary distinct interactions in seed dispersal
networks.** *Oikos* 2022(2):oik.08240. [10.1111/oik.08240](https://doi.org/10.1111/oik.08240)
 19. Ribeiro-Souza P, Graipel ME, Astúa D, Vancine MH, Pires JSR. 2022. **Effects of climate change
on distribution and areas that protect two neotropical marsupials associated with aquatic
environments.** *Ecological Informatics* 68:101570. [10.1016/j.ecoinf.2022.101570](https://doi.org/10.1016/j.ecoinf.2022.101570)
 20. Costa MF, Francisconi AF, Vancine MH, Zucchi MI. 2022. **Climate change impacts on the
Copernicia alba and *Copernicia prunifera* (Arecaceae) distribution in South America.** *Brazilian
Journal of Botany* 45:807-818. [10.1007/s40415-022-00801-8](https://doi.org/10.1007/s40415-022-00801-8)
 21. Muylaert RL, Kingston T, Luo J, Vancine MH, Galli N, Carlson CJ, John RS, Rulli MC, Hay-
man DT. 2022. **Present and future distribution of bat hosts of sarbecoviruses: implica-
tions for conservation and public health.** *Proceedings of the Royal Society B* 289(1975):20220397.
[10.1098/rspb.2022.0397](https://doi.org/10.1098/rspb.2022.0397)
 22. Borges GA, Mancilla G, Siqueira AB, Vancine MH, Ribeiro MC, Maia JCS. 2022. **The fate
of vegetation remnants in the southern Amazons largest threatened hotspot: part (I) a 33-
year analysis of LULCC in the Tapajos River basin, Brazil.** *Research, Society and Development*
11(10):e448111032553. [10.33448/rsd-v11i10.32553](https://doi.org/10.33448/rsd-v11i10.32553)
 23. Galetti M, Carmignotto AP, Percequillo AR, Santos MCO, Ferraz KMPMB, Lima F, Vancine
MH, Muylaert RL, Bonfim FCG, Magioli M, Abra FD, Chiarello AG, Duarte JMB, Morato R,
de Mello Beisiegel B, Olmos F, Galetti Jr. PM, Ribeiro MC. 2022. **Mammals in São Paulo
State: diversity, distribution, ecology, and conservation.** *Biota Neotropica* 22(spe):e20221363.
[10.1590/1676-0611-bn-2022-1363](https://doi.org/10.1590/1676-0611-bn-2022-1363)
 24. Santos PM, Ferraz KMPMB, Ribeiro MC, Niebuhr BB, Vancine MH, Chiarello AC, Paglia AP.
2022. **Natural forest regeneration on anthropized landscapes could overcome climate change
effects on the endangered maned sloth (*Bradypus torquatus*, Illiger 1811).** *Journal of Mammal-
ogy* 103(6):1383-1396. [10.1093/jmammal/gyac084](https://doi.org/10.1093/jmammal/gyac084)
 25. Dutra VAB, Vancine MH, Lima AAM, Toledo PM. 2023. **Dinâmica da paisagem e fragmentação
de ecossistemas em três bacias hidrográficas na Amazônia Oriental entre 1985 e 2019.** *Revista
Brasileira de Geografia Física* 16(02):936-949. [10.26848/rbgf.v16.2.p936-949](https://doi.org/10.26848/rbgf.v16.2.p936-949)
 26. Amaral IS, Pereira JB, Vancine MH, Morales AE, Althoff SL, Gregorin R, Pereira MJR, Valiali
VH, Oliveira LR. 2023. **Where do they live? Predictive geographic distribution of *Tadarida
brasiliensis brasiliensis* (Chiroptera, Molossidae) in South America.** *Neotropical Biology and
Conservation* 18(3):139-156. [10.3897/neotropical.18.e101390](https://doi.org/10.3897/neotropical.18.e101390)
 27. Anunciação PA, Ernst R, Martello F, Vancine MH, Ribeiro MC, Carvalho LMT. 2023. **Climate-
driven loss of taxonomic and functional richness in Atlantic Forest anurans.** *Perspectives in*

28. Teixeira JVS, Bonfim FCG, **Vancine MH**, Ribeiro MC, Oliveira LC. 2023. **Effect of landscape attributes at multiple scales on the occurrence of the threatened golden-headed lion tamarin, *Leontopithecus chrysomelas* Kuhl, 1820 (Primates, Callitrichidae).** *American Journal of Primatology* 86(4):e23588. [10.1002/ajp.23588](https://doi.org/10.1002/ajp.23588)
29. Tonetti V, Bocalini F, Schunck F, **Vancine MH**, Butti M, Ribeiro MC, Pizo M, Balmford A. 2024. **The Protected Areas network may be inefficient to cover biodiversity in a fragmented tropical hotspot under different climate scenarios.** *Perspectives in Ecology and Conservation* 22(1):63-71. [10.1016/j.pecon.2023.12.002](https://doi.org/10.1016/j.pecon.2023.12.002)
30. **Vancine MH**, Muylaert RL, Niebuhr BB, Oshima JEF, Tonetti V, Bernardo R, De Angelo C, Rosa MR, Grohmann CH, Ribeiro MC. 2024. **The Atlantic Forest of South America: spatiotemporal dynamics of remaining vegetation and implications for conservation.** *Biological Conservation* 291:110499. [10.1016/j.biocon.2024.110499](https://doi.org/10.1016/j.biocon.2024.110499)
31. Barbosa FS, **Vancine MH**, Ribeiro MC, Siminski A. 2024. **Análise espacial da fragmentação da paisagem: um estudo de caso sobre os fragmentos florestais destinados à Reserva Legal em duas bacias hidrográficas no Estado do Piauí.** *Revista Observatório de la Economía Latinoamericana* 22(11):e7996. [10.55905/oelv22n11-247](https://doi.org/10.55905/oelv22n11-247)
32. Marques RCM, **Vancine MH**, Suárez YR, Pereira JG, Domingos JD, Silva ABB, Pereira, ZV. 2025. **Dinâmica espaço temporal: variações na composição e configuração da vegetação.** *Revista Brasileira de Geografia Física* 18(2):1334-1348. [10.26848/rbgf.v18.2.p1334-1348](https://doi.org/10.26848/rbgf.v18.2.p1334-1348)
33. Carvalho T, Falconi N, White T, Anjos LA, Giasson LOM, **Vancine MH**, Haddad CFB, Toledo LF, Becker CG. 2025. **The role of seasonal migration in predicting amphibian population persistence across fragmented tropical landscapes: an Individual-Based Model.** *Biodiversity and Conservation* 34:1291-1310. [10.1007/s10531-025-03016-x](https://doi.org/10.1007/s10531-025-03016-x)
34. Gonçalves-Souza T*, **Vancine MH***, Sanders NJ, Haddad NM, (), Chase JM. 2025. **LandFrag: a dataset to investigate the effects of habitat loss and fragmentation on biodiversity in forest fragments.** *Global Ecology and Biogeography [Data Article]* 34(2):e70015. [10.1111/geb.70015](https://doi.org/10.1111/geb.70015)
35. Gonçalves-Souza T, Chase J, Haddad N, **Vancine MH**, Melo FPL, Aizen M, Bernard E, Chiarello GA, Didham R, Faria D, Gibb H, Lima M, Magnago L, Mariano Neto E, Nogueira A, Nemésio A, Passamani M, Pinho BX, Rocha-Santos L, Rodrigues R, Safar N, Santos B, Soto-Werschitz A, Tabarelli M, Uehara-Prado M, Vasconcelos H, Vieira S, Sanders, N. 2025. **Species turnover does not rescue biodiversity in fragmented landscapes.** *Nature* 640:702-706. [10.1038/s41586-025-08688-7](https://doi.org/10.1038/s41586-025-08688-7)
36. Alves-Ferreira G*, **Vancine MH***, Mota FMM, Bello C, Sobral-Souza T, Percequillo AR, Lacher Jr TE, Galetti M, Bovendorp RS. 2025. **From hot to cold spots: climate change will modify endemism centers of small mammals in a biodiversity hotspot.** *Diversity and Distributions* 31(5):e70026. [10.1111/ddi.70026](https://doi.org/10.1111/ddi.70026)
37. Goebel LGA, **Vancine MH**, Bogoni JA, Longo GR, Calicis M, Fearnside PM, Palmeirim AF, Santos-Filho M. 2025. **The impact of Amazon deforestation is magnified by changing the configuration of forest cover.** *Environmental Conservation* 52(3):108-117. [10.1017/S0376892925000086](https://doi.org/10.1017/S0376892925000086)
38. da Silva RFB, Millington JDA, Yue D, **Vancine MH**, Magnago LFS, Viña A, Bin F, Huesca M, Viera SA, Garibaldi LA, Liu J. 2025. **Secondary natural vegetation gains in the Atlantic Forest do not offset losses of carbon stocks and conservation of priority areas.** *Biological Conservation* 312:111512. [10.1016/j.biocon.2025.111512](https://doi.org/10.1016/j.biocon.2025.111512)

39. Francisconi AF, Scaketti M, Morales-Marroquín JA, Fornazier GO, **Vancine MH**, Moro MS, Araujo JVS, Ramos SLF, Modolo VA, Lopes MTG, Zucchi MI. 2025. **Expansion of açai fruit and heart-of-palm production promotes hybridization and introgression in Euterpe palms.** *BMC Plant Biology* 25, 1752. [10.1186/s12870-025-07775-1](https://doi.org/10.1186/s12870-025-07775-1)
40. Gonçalves-Souza T*, **Vancine MH***, Paterno GB*. **Computer vision uncovers trait-based insect responses to habitat loss.** 2026. *Journal of Animal Ecology [Research Highlights]* 95(1), 12-16. [10.1111/1365-2656.70165](https://doi.org/10.1111/1365-2656.70165)
41. **Vancine MH***, Niebuhr BB*, Muylaert RL, Oshima JEF, Tonetti V, Bernardo R, Alves RSC, Zanette EM, Souza VC, Giovanelli JGR, Grohmann CH, Galetti M, Ribeiro MC. 2026. **ATLANTIC SPATIAL: a data set of landscape, topographic, hydrological and anthropogenic metrics for the Atlantic Forest.** *Ecology (Data Paper)* 107(4), e70360. [10.1002/ecy.70360](https://doi.org/10.1002/ecy.70360)
42. **Vancine MH**, Dodonov P, Vilela B, Diele-Viegas L, Souza VC, Nunes FAS, Silva AJO, Milz B, Kita CA, Mello MAR, Muylaert RL. 2026. **The challenges and nuances of teaching the R programming language to ecologists.** *Oecologia Australis*. [link](#)

*Co-first authors

Books

1. Da Silva FR, Gonçalves-Souza T, Paterno GB, Provete DB, **Vancine MH**. 2022. **Análises Ecológicas no R.** Nupeea: Recife, PE, Canal 6: São Paulo. 640 p. ISBN 978-85-7917-564-0. [link](#)

Preprints

1. Muylaert RL, Kingston T, Luo J, **Vancine MH**, Galli N, Carlson CJ, John RS, Rulli MC, Hayman DT. 2021. **Present and future distribution of bat hosts of sarbecoviruses: implications for conservation and public health.** *bioRxiv*. [10.1101/2021.12.09.471691](https://doi.org/10.1101/2021.12.09.471691).
2. **Vancine MH**, Muylaert RL, Niebuhr BB, Oshima JEF, Tonetti V, Bernardo R, De Angelo C, Rosa MR, Grohmann CH, Ribeiro MC. 2023. **The Atlantic Forest of South America: spatiotemporal dynamics of remaining vegetation and implications for conservation.** *bioRxiv*. [10.1101/2023.09.16.558076](https://doi.org/10.1101/2023.09.16.558076).
3. **Vancine MH***, Niebuhr BB*, Muylaert RL, Oshima JEF, Tonetti V, Bernardo R, Alves RSC, Zanette EM, Souza VC, Giovanelli JGR, Grohmann CH, Galetti M, Ribeiro MC. 2023. **ATLANTIC SPATIAL: a data set of landscape, topographic, hydrological and anthropogenic metrics for the Atlantic Forest.** *EcoEvoRxiv*. [10.32942/X26P58](https://doi.org/10.32942/X26P58).

*Co-first authors

For all publications, citations, and journal referees

[ORCID](#), [Web of Science](#), [Scopus](#), [Google Scholar](#)

MANUSCRIPTS UNDER REVIEW

1. Morini RFS*, **Vancine MH***, Martinez AN, Sales LP, Pires MM. **Habitat degradation threatens the topological diversity of seed-dispersal networks in the Atlantic Forest hotspot.** *Global Change Biology*.
2. Furlan ME, Hayman DTS[§], Wada M, Ribeiro MC[§], Muylaert RL[§], **Vancine MH[§]**. **Optimizing human rabies prevention in Brazil through spatial prioritization and a One Health approach.** *Scientific Reports*.

3. Amaral IS, Bandeira-Pereira J, Vancine MH, Althoff SL, Gregorin R, Pereira MJR, Valiali VH, Oliveira LR. **Predictive distribution of the Brazilian free-tailed bat *Tadarida brasiliensis brasiliensis* (Chiroptera, Molossidae) under climate change scenarios.** *Biological Conservation*.
4. Amaral IS, Bandeira-Pereira J, Vancine MH, Althoff SL, Gregorin R, Pereira MJR, Valiali VH, Oliveira LR. **Glacial refugia and environmental suitability shaped the historical distribution of *Tadarida brasiliensis brasiliensis* (Chiroptera, Molossidae) in South America.** *Acta Oecological*.
5. Redana M, Moura MR, Starinchak B, Vancine MH, Brando P, Goetz S, Jantz P. **Forest structure types in the southern Amazon Basin are influenced by anthropogenic edges and physiography.** *Remote Sensing of Environment*.
6. Martinez AN, Vancine MH, Pizo MA, Silva WR, Guimarães Jr. PR, Keitt TH, Pires MM. **Estimating landscape connectivity for plants through the lens of seed dispersers.** *Journal of Ecology*.

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§Co-senior authors

TEACHING EXPERIENCE

Graduate courses (450 h)

- **Ecological Niche Modeling in R (30 h)** 2016
Teaching assistant, Ecology and Biodiversity Graduate Program, São Paulo State University (UNESP)
- **Ecological Niche Modeling: theory and practice (45h)** 2016
Teaching assistant, Ecology Graduate Program, State University of Campinas (Unicamp)
- **Ecological Niche Modeling: theory and practice (60 h)** 2017
Teaching assistant, Ecology and Biodiversity Graduate Program, São Paulo State University (UNESP)
- **Introduction to Geoprocessing for Ethnobiology and Conservation (45 h)** 2019
Invited teacher, Ethnobiology and Nature Conservation Graduate Program, Rural Federal University of Pernambuco (UFRPE)
- **Introduction to the use of geospatial data in R (60 h)** 2020
Invited teacher, Ecology, Biodiversity and Evolution Graduate Program, São Paulo State University (UNESP)
- **Introduction to the use of geospatial data in R (60 h)** 2021
Invited teacher, Ecology, Biodiversity and Evolution Graduate Program, São Paulo State University (UNESP)
- **NT265/NE441 - Data analysis and graphical presentation using R language (60 h)** 2024
Invited teacher, Ecology Graduate Program, State University of Campinas (Unicamp)
- **Version control with git/GitHub and introduction to programming in R (30 h)** 2025
Invited teacher, Ecology, Biodiversity and Evolution Graduate Program, Federal University of Paraíba (UFPB)
- **NT265/NE441 - Data analysis and graphical presentation using R language (60 h)** 2025
Invited teacher, Ecology Graduate Program, State University of Campinas (Unicamp)

Undergraduate courses (102 h)

- **Quantitative Ecology (60 h)** 2015
Teaching assistant, Ecology Undergraduate, São Paulo State University (UNESP)
- **Statistical Models in Ecology (12 h)** 2018
Teacher, Ecology Undergraduate, São Paulo State University (UNESP)
- **Statistical Models in Ecology (30 h)** 2020
Teacher, Ecology Undergraduate, São Paulo State University (UNESP)

Workshops (116 h)

- **Organization of data in electronic sheets - Calc (08 h)** 2014
Workshop instructor, XXV Ecology Studies Week, São Paulo State University (UNESP)
- **Introduction to software R (08 h)** 2015
Workshop instructor, XXVI Ecology Studies Week, São Paulo State University (UNESP)
- **Field herpetology (15 h)** 2016
Workshop instructor, XXVII Ecology Studies Week, São Paulo State University (UNESP)
- **Introduction to software R (16 h)** 2016
Workshop instructor, XXVII Ecology Studies Week, São Paulo State University (UNESP)
- **Field work with amphibians (08 h)** 2018
Workshop instructor, 29th Biology Studies Week, São Paulo State University (UNESP)
- **Introduction to Species Distribution Modeling using R Language: theory and practice (07 h)** 2019
Workshop instructor, 9th Brazilian Congress of Herpetology, State University of Campinas (Unicamp)
- **Introduction to R Language (08 h)** 2019
Workshop instructor, XXX Ecology Studies Week, São Paulo State University (UNESP)
- **Introduction to R Language (08 h)** 2019
Workshop instructor, 30th Biology Studies Week, São Paulo State University (UNESP)
- **Introduction to version control with Git, GitHub, and RStudio (04 h)** 2021
Workshop instructor, Rural Federal University of Pernambuco (UFRPE)
- **Geoprocessing with QGIS (03 h)** 2021
Workshop instructor, 32nd Biology Studies Week, São Paulo State University (UNESP)
- **Introduction to species distribution modeling using the R language (06 h)** 2021
Workshop instructor, Mastozólogos Organizados em uma Conferência Online (MOCÓ), Sociedade Brasileira de Mastozoologia
- **Introduction to the R language for data manipulation and visualization (08 h)** 2022
Workshop instructor, XXXII Ecology Studies Week, São Paulo State University (UNESP)
- **Analyzing spatial patterns in biodiversity data (06 h)** 2024
Workshop instructor, São Paulo School of Advanced Science (SPSAS), State University of Campinas (Unicamp)

- **Introduction to landscape metrics in R (02 h)** 2024
Workshop instructor, I Brazilian Symposium on Ecological Synthesis, Federal University of Santa Maria (UFSM)

Guest lectures (04 h)

- **Introduction to the R language (02 h)** 2023
Teacher, Statistical Models in Ecology, São Paulo State University (UNESP)
- **Introduction to Numerical Ecology (02 h)** 2023
Teacher, Quantitative Ecology, São Paulo State University (UNESP)

ADMINISTRATIVE EXPERIENCE

- **Student representative** 2020-2021
Graduate Program in Ecology, Evolution and Biodiversity, São Paulo State University (UNESP)

PROFESSIONAL AFFILIATIONS AND MEMBERSHIPS

- Associação Brasileira de Ecólogos (ABE) 2018-Present
- Associação Brasileira de Ciência Ecológica e Conservação (ABECO) 2021-Present
- Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology (SORTEE) 2025-Present

STUDENT SUPERVISION

- **Helena Thereza Carvalho de Oliveira** 2021
Distribuição do padrão reprodutivo em comunidades de anuros na Mata Atlântica. [link](#)
Advisor, Undergraduate thesis, Biological Sciences, São Paulo State University (UNESP)
- **Bruno Eduardo Ribeiro Silva** 2022
Comparação da distribuição de espécies inferidos por Modelos de Nicho Ecológicos e mapa de especialistas da IUCN para anfíbios anuros da América do Sul. [link](#)
Advisor, Undergraduate thesis, Ecology, São Paulo State University (UNESP)
- **Maria Eduarda Furlan** 2025
Inferindo a circulação do vírus da raiva e risco zoonótico com aplicações para a saúde pública. [link](#)
Advisor, Undergraduate thesis, Ecology, São Paulo State University (UNESP)
- **Mariana Trindade Lahr** 2025
Movimento e mortalidade: efeito das rodovias sobre a mastofauna e conservação ambiental no Brasil. [link](#)
Advisor, Undergraduate thesis, Ecology, São Paulo State University (UNESP)

MEETINGS AND PRESENTATIONS

Conferences

- Brazilian Amphibian Conservation Symposium (ANFoCO) 2018
- IX Brazilian Congress of Herpetology 2019
- II Workshop of Community Ecology 2020
- Mastozoologists Organized in an Online Conference 2021
- FOSS4G (Free and Open Source Software for Geospatial) 2021

- I Brazilian Symposium on Ecological Synthesis 2024
- ATBC 2025 2025
- I MapBiomas User Summit 2025
- 2026 Lemann Dialogue: Six Biomes, Multiple Realities 2026
- EEB 25th Anniversary Science Symposium 2026

SERVICES

- Journal editor:
Austral Ecology
- Journal reviewer:
Zoologia, PLOS One, Biological Invasions, Hydrobiologia, Scientific Data, Diversity and Distributions, Papéis Avulsos de Zoologia, Conservation Biology, Austral Ecology, Scientia Plena, Ecosystem Services, PeerJ, Global Change Biology, Journal of Biogeography, Biotropica, Ocean and Coastal Management, Journal of Applied Ecology, Global Ecology and Biogeography, Ecological Indicators, Journal of Animal Ecology, Biological Conservation, Ecosphere, Ecological Modelling, Basic and Applied Ecology, Landscape Ecology, Ecological Applications, Restoration Ecology, Perspectives in Ecology and Conservation, Regional Environmental Change.
- Funding agency referee:
São Paulo Research Foundation (FAPESP)

SKILLS

Programming

- R (advanced, including tidyverse, markdown, quarto), Shiny (basic), git (intermediate), LaTeX (intermediate), Python (basic), Julia (basic), JavaScript (basic), shell/bash (basic), HTML/CSS (basic)

Software, applications and platforms

- QGIS (advanced), GRASS GIS (advanced), GNU/Linux (intermediate), Fragstats (intermediate), GuidosToolbox (intermediate), LibreOffice suite (intermediate), Zotero (intermediate), Google Earth Engine (basic), GeoDa (basic), ArcGIS (basic), Inkscape (basic), Sublime Text (basic), Emacs (basic)

R packages

- *lsmetrics*: provides multiscale calculation and spatialization of landscape metrics using GRASS GIS
Authors: Bernardo B. Niebuhr*, **Maurício Humberto Vancine***, Felipe Martello Ribeiro, Renata L. Muylaert, John Wesley Ribeiro, Milton Cezar Ribeiro
Link: mauriciovancine.github.io/lsmetrics
- *atlanticr*: provides data from the ATLANTIC, data papers from Atlantic Forest
Authors: **Maurício Humberto Vancine**, Bernardo Niebuhr, Renata L. Muylaert, Mauro Galetti, Milton Cezar Ribeiro
Link: mauriciovancine.github.io/atlanticr
- *amphiBR*: dataset from the official publication of the List of Amphibians in Brazil published by the Brazilian Society of Herpetology

Authors: Paulo Barros de Abreu Junior, **Maurício Humberto Vancine**, Diogo B. Provete

Link: paulobarros.github.io/amphiBR

- *ecodados*: ecological data base for teaching statistics

Authors: Gustavo Paterno, Diogo B. Provete, Fernando Rodrigues da Silva, Thiago Gonçalves-Souza, **Maurício Humberto Vancine**

Link: paternogbc.github.io/ecodados

- *lsma*: landscape-structure multiscale analysis using R

Authors: Wilson Frantini-Silva, Lazaro S. Carneiro, André Luis Regolin, **Maurício Humberto Vancine**, Juliana S. Santos, Bernardo Brandão Niebuhr, Edson Valmorbida, Maria Cristina Gaglianone, Milton C. Ribeiro

Link: wilsonfrantini/lsma

*Co-first authors

Statistical knowledge

- *Descriptive statistics*: handling tabular data (advanced), handling spatial data (advanced), tabular data visualization (advanced), spatial data visualization (advanced).
- *Univariate statistics*: frequentist methods (advanced), likelihood-based methods (intermediate), model selection (intermediate), Bayesian inference (basic).
- *Multivariate statistics*: ordination methods (intermediate), clustering methods (intermediate), discriminant analysis (basic).

Languages

- English (intermediate)
- Spanish (basic)
- Italian (basic)
- Portuguese (native)

MEDIA

News

- **Pesquisadores mapeiam áreas com potencial para futuros surtos de doenças semelhantes à Covid-19.** *Jornal da UNESP.* [link](#) 2022
- **Alto Tietê somou mais de 30 hectares desmatados em quatro anos.** *G1.* [link](#) 2023
- **Novo estudo destaca preocupação com estado de conservação da Mata Atlântica.** *Um Só Planeta.* [link](#) 2024
- **Novo estudo destaca preocupação com estado de conservação da Mata Atlântica.** *Folha de São Paulo.* [link](#) 2024
- **Novo estudo destaca preocupação com estado de conservação da Mata Atlântica.** *Agência FAPESP.* [link](#) 2024
- **Novo estudo destaca preocupação com estado de conservação da Mata Atlântica.** *Jornal da UNESP.* [link](#) 2024

- **Condomínios, estradas e caça: Mata Atlântica enfrenta novos riscos.** *Olhar Digital [Ciência e Espaço]*. [link](#) 2024
- **Desmatamento desacelera na Mata Atlântica, mas fragmentação aumenta.** *Mongabay Brasil*. [link](#) 2024
- **Unidades de Conservação: guardiãs da Mata Atlântica.** *Ekos Brasil*. [link](#) 2024
- **Consciência ambiental é aceleradora de projetos de restauração na Mata Atlântica.** *Um Só Planeta*. [link](#) 2024
- **Seleção Natural Platform: Defining Contributions to Biodiversity Conservation in Natural Environments.** *Land Innovation Foundation*. [link](#). 2024
- **Plataforma de Selección Natural: Definiendo Contribuciones a la Conservación de la Biodiversidad en Ambientes Naturales.** *Land Innovation Foundation*. [link](#). 2024
- **No Clima dos Biomas: com restauração de florestas e nascentes de água, projetos buscam recuperar uma Mata Atlântica degradada.** *Um Só Planeta*. [link](#) 2024
- **O paradoxo da Mata Atlântica.** *Piauí*. [link](#) 2025
- **Fragmented Forests Lead to Declining Biodiversity.** *ClimateAction*. [link](#) 2025
- **Manchas grandes de floresta têm mais biodiversidade do que fragmentos menores somados.** *Revista Fapesp*. [link](#) 2025
- **Fragmentação de florestas reduz biodiversidade em todas as escalas, mostra estudo global.** *The Conversation*. [link](#) 2025
- **Preservar a biodiversidade exige áreas de conservação amplas.** *EcoDebate*. [link](#) 2025
- **Estudo da Nature analisa mais de 4.000 espécies e mostra que é preciso grandes áreas de floresta para preservar diversidade.** *bori agência*. [link](#) 2025
- **Fragmentação florestal e perda de habitat prejudica a biodiversidade.** *Correio Braziliense*. [link](#) 2025
- **Pequenas florestas isoladas não chegam para proteger a biodiversidade.** *Greensavers*. [link](#) 2025
- **Want to preserve biodiversity? Go big, U-M researchers say.** *Michigan News*. [link](#) 2025
- **New study refutes habitat fragmentation claim.** *IDiv*. [link](#) 2025
- **Resolving a 50-year debate: fragmentation decreases biodiversity on multiple scales.** *Conservation Corridor*. [link](#) 2025
- **New paper resolves a half-century old debate about how to conserve biodiversity in natural areas.** *Anthropocene*. [link](#) 2025
- **Em conservação, tamanho é mesmo documento.** *oeco*. [link](#) 2025
- **Conservar a biodiversidade exige grandes áreas florestais e paisagens contínuas.** *Jornal da USP*. [link](#) 2025
- **Mudanças climáticas podem eliminar pequenos mamíferos únicos da Mata Atlântica.** *CDS Litoral Sul*. [link](#) 2025
- **Mudanças Climáticas Podem Eliminar Pequenos Mamíferos Únicos da Mata Atlântica da América do Sul.** *UESC Notícias*. [link](#) 2025

- O que é melhor proteger: grandes florestas ou vários fragmentos? Novo estudo reacende polêmica. *Mongabay*. [link](#) 2025
- Protect one large forest, or many small ones? New study reignites conservation debate. *Mongabay*. [link](#) 2025
- Ganhos de vegetação secundária na Mata Atlântica são insuficientes para compensar perdas de estoque de carbono e áreas de conservação. *UFSB Ciência*. [link](#) 2025
- Ensinando programação para quem fugiu da matemática. *Blog InfloresCiência*. [link](#) 2026

YouTube

- Métricas de paisagem no R. *GeoCast Brasil*. [link](#) 2019
- We R Live 01: Como começar no R. *GeoCast Brasil*. [link](#) 2020
- We R Live 02: Elaborando mapas no R. *GeoCast Brasil*. [link](#) 2020
- We R Live 03: Elaborando mapa no R com tmap. *GeoCast Brasil*. [link](#) 2020
- We R Live 04: Manipulando dados raster no R. *GeoCast Brasil*. [link](#) 2020
- We R Live 05: Manipulando dados raster no R II. *GeoCast Brasil*. [link](#) 2020
- We R Live 06: Extraíndo dados climáticos para pontos. *GeoCast Brasil*. [link](#) 2020
- We R Live 07: Introdução à estatística Espacial I. *GeoCast Brasil*. [link](#) 2020
- We R Live 08: Introdução à Estatística Espacial II. *GeoCast Brasil*. [link](#) 2020
- We R Live 09: Introdução à estatística Espacial III. *GeoCast Brasil*. [link](#) 2020
- We R Live 10: Join como associar dados à vetores. *GeoCast Brasil*. [link](#) 2020
- We R Live 11: florestal - Pacote R para inventário florestal. *GeoCast Brasil*. [link](#) 2020
- We R Live 13: Introdução à Estatística Espacial IV. *GeoCast Brasil*. [link](#) 2020
- We R Live 14: Geoestatística com Jorge Kazuo Yamamoto. *GeoCast Brasil*. [link](#) 2020
- We R Live 15: Introdução à estatística Espacial V. *GeoCast Brasil*. [link](#) 2020
- We R Live 16: R Markdown: usando o R para comunicar seus resultados. *GeoCast Brasil*. [link](#) 2020
- Uso e aplicações de modelos estatísticos em Ecologia e Geografia. *EstaTiDados*. [link](#) 2020
- Aplicações da Cartografia para a Ecologia Espacial. *Ciclo de Palestras de Cartografia Básica (UFF)*. [link](#) 2021
- Estatística e análise de dados espaciais no R: um estudo de caso com dados do Lago Walker. *GeoCast Brasil*. [link](#) 2021
- Conhecendo Hugo Apéro: fazendo blogs com blogdown e R. *GeoCast Brasil*. [link](#) 2021
- O fascinante mundo do GRASS GIS. *GeoCast Brasil*. [link](#) 2021
- O fascinante mundo do GRASS. *Fascinante Mundo do Sensoriamento Remoto*. [link](#) 2021
- Modelos em Ecologia: extrapolando nosso conhecimento sobre a distribuição das espécies. *DEA UFV*. [link](#) 2021

- **Análises Ecológicas no R (Lançamento).** *Análises Ecológicas no R.* [link](#) 2022
- **Análise de dados geoespaciais no R.** *Grupo de Estudos em Ecologia Quantitativa (GEEQ).* [link](#) 2024
- **Análises Ecológicas no R e a importância das análises estatísticas para a ecologia e a transdisciplinaridade.** *Engenharia Ambiental Ibirité UEMG.* [link](#) 2024

Podcasts and interviews

- **Unesp realiza estudo para verificar a importância do isolamento social.** *TV Claret.* [link](#) 2020
- **O Fascinante Mundo do Sensoriamento Remoto.** *GeoCast Brasil.* [link](#) 2020
- **Conversa com Ecólogo II - A importância da análise de dados.** *Associação Brasileira de Ecólogos (ABE) - Conversa com Ecólogo.* [link](#) 2020
- **Você tem medo de errar?** *DesAbrçando Árvores.* [link](#) 2022
- **Pesquisa da Unesp aponta melhora na conservação da Mata Atlântica.** *RecordTV Interior SP.* [link](#) 2024
- **Análises Ecológicas no R (Eng. Ambiental e Biologia).** *Bate-papo com cientistas - UEMG Ibiraté.* [link](#) 2024

REFERENCES

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